Microcalculator

«Электроника МК-59»

(Elektronika MK-59)

Operating Manual

Руководство по эксплуатации

English edition

Manufacturer: Rodon (formerly Positron) plant, Ivano-Frankivsk

About this edition. This is an English rendering of the factory operating manual, made from a scan of a copy sold in 1987. It follows the original's numbered sections, tables and worked examples. The purchase checks, warranty terms and tear-off coupons (sections 1 and 10) are summarised, since they applied to the original Soviet retail sale. Translator's notes are marked **[TN]**. The plant emblem on this cover (in use 1977–1994) is from Wikimedia Commons (“Родон Ивано-Франковск эмблема 1972-1994.svg”), public domain.

[TN] Key labels. This manual is printed for the earlier key legends. Later calculators (and the circuit diagram) use different labels for four keys, in the same positions:

This manual	Later calculators	Function
Ф	**РФЕ**	Fixed decimal point
Е	**РЕЖ**	Natural (floating) decimal point
СК	**СХ**	Clear the keyboard (entry) register
ИП	**П→Х**	Recall memory to the display

1. Attention!

1.1. When buying the calculator, the purchaser was to have it checked against the control test (Table 1, see the appendix), check that the set was complete, and make sure the warranty and tear-off coupons carried the shop's stamp, the seller's signature or stamp and the date of sale. The number on the warranty coupon had to match the number on the base of the calculator, and the seal on the case had to be intact. *(Summarised.)*

1.2. After storage in a cold room, or after transport in winter, keep the calculator at room temperature for **8 hours** before switching it on.

1.3. Read this manual carefully before using the calculator.

2. Contents of the set

Item	Quantity
Microcalculator «Электроника МК-59»	1
Operating manual	1
Packaging	1 set

3. Safety instructions

3.1. If the calculator develops a fault, switch it off and disconnect it from the mains.

3.2. Do not open the calculator, and do not carry out any work on it while it is switched on. The calculator may be opened and repaired only by persons authorised to repair it.

4. Brief description

4.1. Purpose

4.1.1. The «Электроника МК-59» is a desktop calculator for individual use. It is intended for a wide range of planning and economic, accounting and statistical, bookkeeping and household calculations.

4.2. The «Электроника МК-59» is a 16-digit calculator with a natural representation of the decimal point. When required, results can be given with a fixed decimal point, with the number of decimal places selectable from 0 to 9. It automatically performs arithmetic with constants, storage and accumulation of data, and rounding.

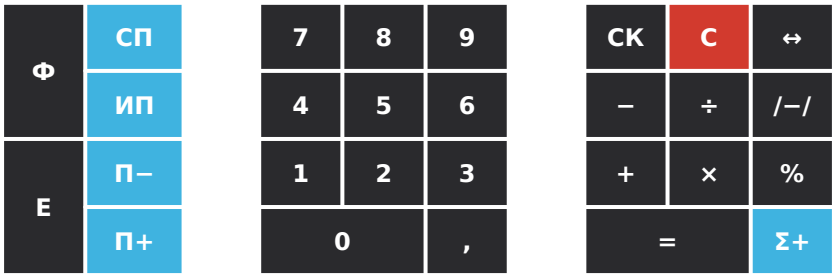
Numbers, operations and commands are entered by pressing the corresponding keys. Input data and results are checked visually on the 16-digit display.

A memory register is provided for storing and accumulating intermediate results.

The display indicates the sign of a negative number, that the memory register is in use, and overflow of the calculator's digit capacity.

5. Keyboard

Layout of the keys (as printed in this manual):



Data entry keys

Key	Function
0 … 9	Enter numeric data.
,	Enter the decimal point (a comma in Russian practice).

Operation keys

Key	Function
+	Carries out the operation set up before (if there is one); the calculator then prepares to add the number on the display to the number keyed in next. The result is obtained by pressing any of + − × ÷ or =.
−	Carries out the operation set up before (if there is one); the calculator then prepares to subtract the number keyed in next from the number that was on the display when − was pressed. The result is obtained by pressing any of + − × ÷ or =.
×	Carries out the operation set up before (if there is one); the calculator then prepares to multiply two numbers, one of which is on the display and the other entered from the keyboard. The result is obtained by pressing any of + − × ÷ or =.
÷	Carries out the operation set up before (if there is one); the calculator then prepares to divide the number on the display by the number entered from the keyboard. The result is obtained by pressing any of + − × ÷ or =.
=	Carries out the operation prepared before.
/−/	Changes the sign of the number on the display.
↔	The number on the display is replaced by the number held in the working register (exchanges the two operands).
%	Calculates a percentage of a number.
П+	Enters the number into the memory register and adds it to the number stored there.
П−	Enters the number into the memory register and subtracts it from the number stored there.
СП	Clears the memory register. **[TN]** The worked examples (Table 2, items 10.1 and 10.3) show that it first displays the memory contents as a total, then clears it.
ИП	Recalls the contents of the memory register to the display (memory is kept).
СК	Resets the keyboard register (clears the number being entered).
С	Resets the operating registers.
Ф	Switches the calculator to giving results with a fixed decimal point. **[TN]** Follow it with a digit 0–9 for the number of decimal places (see Table 2).
Е	Switches the calculator to giving results with a natural (floating) decimal point.
Σ+	Automatically accumulates the results of calculations in the operating registers into the memory register.

6. Technical data

Parameter	Value
Operations performed	Addition, subtraction, division, multiplication, change of sign, calculation of percentages, percentage ratios, discounts and markups; operations with percentages and constants; exchange of the contents of the operating registers.
Operations with the memory register	Addition and subtraction; recall from memory keeping the contents (intermediate result); recall from memory clearing the contents (intermediate or final result); automatic accumulation of results; rounding of the result in fixed-point mode.
Decimal point, on input	Natural
Decimal point, on output	Natural, or fixed with the number of decimal places selectable from 0 to 9
Number of memory registers	1
Characters displayed	0–9, minus sign, decimal comma, and a point (service indicator)
Average time per operation	no more than 0.35 s
Display capacity	16 digits
Range of numbers	10^{-14} to $10^{16} - 1$
Number system, input and output	Decimal
Power consumption	no more than 8 W
Supply	220 V (+10 % / −15 %), 50 ± 1 Hz
Operating temperature	+10 to +35 °C
Dimensions	236 × 223 × 70 mm
Mass	no more than 1.5 kg

7. Preparing for use

7.1. Before plugging the calculator into the mains, check the position of the power switch. It must be off: the red dot on the front end of the switch must not be visible.

7.2. Plug into the mains socket and set the power switch to ON (the red dot on the front end of the switch should now be visible). Then press the keys **Е**, **СП**, **С** in turn.

7.3. The calculator is ready for work when the digit 0 appears in the lowest (rightmost) display position within 10 seconds of pressing those keys.

8. Operating procedure

8.1. Examples of calculations are given in Table 2. Key presses are shown as key caps; numbers are typed on the digit keys. “Dec.” is the number of decimal places selected; “Mem.” marks examples in which the service position shows that memory is in use.

Table 2. Worked examples

Calculation	Dec.	Mem.	Key sequence	Display
1. Bring the calculator to its initial state and set the decimal point to the required position (e.g. two places in fixed-point mode)	2		**СП** **С** **Ф** 2	0 0,00
2. Addition and subtraction: 13,56 − 65,21 + 85,9	2		**Ф** 2 13,56 **−** 65,21 **+** 85,9 **=**	0,00 34,25
Multiplication: 12,34 × 4,56 **[TN: see note]**	2		**Ф** 2 12,34 **×** 4,65 **=**	57,38
Division: 123,4 : 45,675	2		**Ф** 2 123,4 **÷** 45,675 **=**	2,70
3. Switching to results with a natural decimal point			**Е**	2,701696770 66228
4. Operations with constants 4.1. 1,28 × 4,56; 7,89 × 4,56 Operations with a constant divisor, addend or subtrahend are done in the same way.			1,28 **×** 4,56 **=** 7,89 **=**	5,8368 35,9784
4.2. 360 : 24; 360 : 45 Operations with a constant minuend are done in the same way.		•	360 **П+** **÷** 24 **=** **ИП** **÷** 45 **=**	15 8
5. Change of sign: 6 → −6			6 **/−/**	−6
6. Exchanging the contents of the operating registers 6.1. 21,6 / (5,1 + 3,7)	4		**Ф** 4 5,1 **+** 3,7 **÷** 21,6 **↔** **=**	2,4546
6.2. 56,8 / ((6,2 − 3,48) × 9,5 − 5,84)	2		**Ф** 2 6,2 **−** 3,48 **×** 9,5 **−** 5,84 **÷** 56,8 **↔** **=**	2,84
7. Percentages: 12 % of 48			12 **×** 48 **%**	5,76
8. Percentage ratios 8.1. 5,76 / 48 × 100 %			5,76 **÷** 48 **%**	12
8.2. (1000 − 800) / 800 × 100 %			1000 **−** 800 **÷** 800 **%**	25
8.3. (1000 − 800) / 1000 × 100 %		•	**СП** 1000 **П+** **−** 800 **÷** **ИП** **%**	20
9. Markup and discount 125 + (3 % of 125) 125 − (3 % of 125)			125 **+** 3 **%** 125 **−** 3 **%**	128,75 121,25
10. Automatic accumulation 10.1. 8 : 3 + 4 : 3 − 5 × 6	3	•	**Ф** 3 8 **÷** 3 **Σ+** 4 **Σ+** 5 **/−/** **×** 6 **Σ+** **СП**	−26
10.2. 27300 / (120 + 6 + (140 × 0,05) + 140) + 5460 / (120 + 6 + (140 × 0,05) + 140) + 600		•	**СП** 120 **П+** 6 **П+** 140 **×** 0,05 **+** 140 **Σ+** 27300 **÷** **СП** **Σ+** 5460 **Σ+** 600 **+** **ИП** **Σ+**	720
10.3. 125 + 30 % of 125 145 + 25 % of 145 165 + 40 % of 165 Total:		•	**СП** 125 **+** 30 **%** **П+** 145 **+** 25 **%** **П+** 165 **+** 40 **%** **П+** **СП**	162,5 181,25 231 574,75
11. Raising to a power 13^4 2^{-2}			1 **×** 13 **=** **=** **=** **=** 1 **÷** 2 **=** **=**	28561 0,25

Calculation	Dec.	Mem.	Key sequence	Display
12. Value of a deposit after 4 years with 3 % annual growth and an initial deposit of 1000: $1000 \times (1 + 0{,}03)^4$	2		**Ф** 2 1 **×** 1,03 **=** **=** **=** **=** **×** 1000 **=**	1125,51
13. First payment on a loan of 8000 to be repaid in three annual instalments at 6 % a year: $8000 \times 0{,}06 / (1 - 1 / (1 + 0{,}06)^3)$	2		**Ф** 2 1 **×** 1,06 **=** **=** **=** **÷** 1 **↔** **+** **/−/** 1 **÷** 0,06 **↔** **×** 8000 **=**	2992,88

[TN] In example 2 the heading says 12,34 × 4,56 but the key sequence enters 4,65; the printed result 57,38 matches 12,34 × 4,65. This is a misprint in the original. In examples 12 and 13, pressing **=** repeatedly repeats the last multiplication, which is how powers are formed.

Notes

1. When the digit capacity overflows, a point lights at the **top** of the leftmost service position and the calculator locks. To unlock it press **C**; the calculator returns to its initial state and the contents of the memory register are unchanged.

If the overflow occurred in the memory register, press **СК** and **C**; the calculator returns to its initial state. To clear the memory register press **СП**.

2. When the memory register holds information, a point lights at the **bottom** of the leftmost service position of the display.

9. Transport and storage

9.1. During transport the calculator must be protected from climatic effects and mechanical damage.

9.2. When sent for repair, the calculator must be packed in a transport container that prevents it from moving about inside, and protects it from mechanical damage, dust, moisture and climatic effects.

9.3. Store the calculator in a dry, heated room, free of acid, alkali and other aggressive vapours, at +10 to +35 °C and a relative humidity of no more than 80 %.

9.4. Storage life: 2 years from the date of manufacture.

10. Warranty (summary)

- The manufacturer guaranteed the calculator against its technical specifications for **24 months** from the date of sale (or from the date of manufacture if the coupons were not stamped by the shop), provided the conditions of use, transport and storage were observed.
- Faults within the warranty period were repaired free of charge by the repair enterprise serving the owner's district. Exchange followed the Soviet rules for exchanging manufactured goods (1974).
- Calculators were not accepted for warranty repair without a completed coupon, or with a broken factory seal, dents, signs of impact or other mechanical damage.
- The original ends with the warranty coupon and tear-off service coupons. The scanned copy was sold in 1987 at a price of 125 roubles.

Appendix – Table 1: Test program for checking the calculator

This self-test was used to check the calculator when it was sold. Press the keys one row at a time and compare the display with the expected reading. Flags: **Dec.** = decimal places selected; **Mem.** = memory dot (bottom of the leftmost position); **−** = minus sign; **Ovf.** = overflow dot (top of the leftmost position).

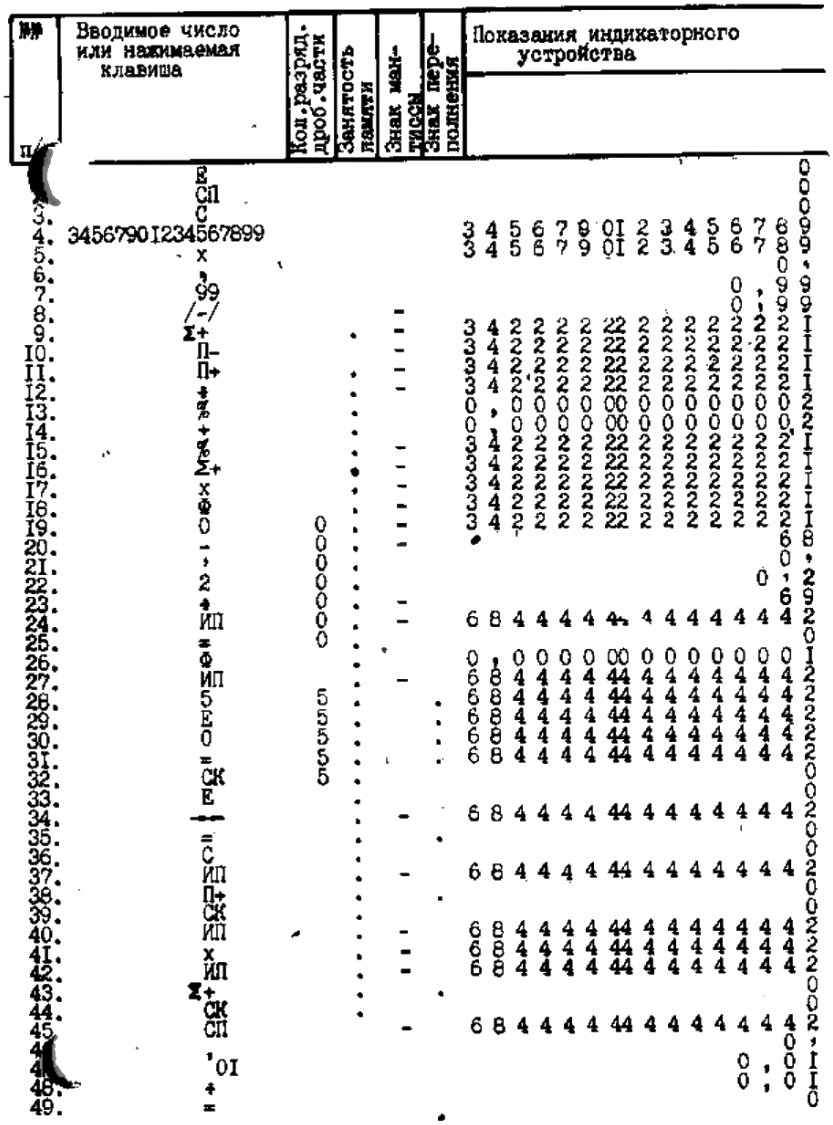
[TN] Transcribed from the scan, with the keys and every reading confirmed by running the test on a working MK-59. A blank flag column means no mark is shown. The original scan follows the table for comparison.

No.	Manual key	Later key	Dec.	Mem.	−	Ovf.	Expected reading
1	**Е**	**РЕЖ**					0
2	**СП**	**СП**					0
3	**С**	**С**					0
4	**34567901234567899**	**same**					3456790123456789
5	**×**	**×**					3456790123456789
6	**,**	**,**					0,
7	**99**	**99**					0,99
8	**/−/**	**/−/**			−		−0,99
9	**Σ+**	**Σ+**		•	−		3422222222222221
10	**П−**	**П−**			−		3422222222222221
11	**П+**	**П+**		•	−		3422222222222221
12	**÷**	**÷**		•	−		3422222222222221
13	**%**	**%**		•			0,00000000000002
14	**+**	**+**		•			0,00000000000002
15	**%**	**%**		•	−		3422222222222221
16	**Σ+**	**Σ+**		•	−		3422222222222221
17	**×**	**×**		•	−		3422222222222221
18	**Ф**	**РФЕ**		•	−		3422222222222221
19	**0**	**0**	0	•	−		3422222222222221
20	**−**	**−**	0	•	−		68 (dot at far left)
21	**,**	**,**	0	•			0,
22	**2**	**2**	0	•			0,2
23	**÷**	**÷**	0	•	−		69
24	**ИП**	**П→Х**	0	•	−		6844444444444442
25	**=**	**=**	0	•			0
26	**Ф**	**РФЕ**		•			0,00000000000001
27	**ИП**	**П→Х**		•	−		6844444444444442
28	**5**	**5**	5	•		•	6844444444444442
29	**Е**	**РЕЖ**	5	•		•	6844444444444442
30	**0**	**0**	5	•		•	6844444444444442
31	**=**	**=**	5	•		•	6844444444444442
32	**СК**	**СХ**	5	•			0
33	**Е**	**РЕЖ**		•			0
34	**↔**	**↔**		•	−		6844444444444442
35	**=**	**=**		•		•	0
36	**С**	**С**		•			0
37	**ИП**	**П→Х**		•	−		6844444444444442

No.	Manual key	Later key	Dec.	Mem.	−	Ovf.	Expected reading
38	**П+**	**П+**		•		•	0
39	**СК**	**СХ**		•			0
40	**ИП**	**П→Х**		•	−		6844444444444442
41	**×**	**×**		•	−		6844444444444442
42	**ИП**	**П→Х**		•	−		6844444444444442
43	**Σ+**	**Σ+**		•		•	0
44	**СК**	**СХ**		•			0
45	**СП**	**СП**			−		6844444444444442
46	**,**	**,**					0,
47	**01**	**01**					0,01
48	**÷**	**÷**					0,01
49	**=**	**=**				•	0

[TN] The decimal comma takes a whole display position, so a number shows at most 16 digits, or 15 digits plus the comma; 0,00000000000001 (step 26) is the smallest value that fits. Step 4 enters 17 digits; only the first 16 are accepted, so the display shows 3456790123456789. Step 9 multiplies by −0,99: 3456790123456789 × 0,99 = 3422222222222221,11, shown as an integer because the display is full. 6844444444444442 is twice that value.

Table 1 as printed in the original (for comparison)



№ п/п	Вводимое число или нажимаемая клавиша	Кол. разряд. дроб. части	Занятость памяти	Знак мантиссы	Знак перенолнения	Показания индикаторного устройства
1.	В↑					0
2.	СП					0
3.	С					0
4.	3456790I234567899					3 4 5 6 7 9 0I 2 3 4 5 6 7 8 9
5.	×					3 4 5 6 7 9 0I 2 3 4 5 6 7 8 9
6.	,					0
7.	99					0 , 9 9
8.	/-/			-		0 , 9 9
9.	M+			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
10.	П-			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
11.	П+			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
12.	4			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
13.	0			-		0 , 0 0 0 0 0 00 0 0 0 0 0 0 0 2
14.	0			-		0 , 0 0 0 0 0 00 0 0 0 0 0 0 0 2
15.	4			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
16.	4			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
17.	4			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
18.	4			-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 2 2
19.	0	0		-		3 4 2 2 2 2 2 22 2 2 2 2 2 2 6 8
20.	.	0		-		• 6 0 ,
21.	2	0		-		0 , 2 9
22.	+	0		-		6 2
23.	ИП	0		-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
24.	e	0		-		0 , 0 0 0 0 0 00 0 0 0 0 0 0 0 1
25.	ИП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
26.	5	5		-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
27.	0	5		-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
28.	ВБР	5		-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
29.	ВОЗ	5		-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
30.	ВЗК	5		-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
31.	ВП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
32.	С/П			-		0
33.	ВП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
34.	ИП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
35.	ВП			-		0
36.	ИП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
37.	ВП			-		0
38.	ИП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
39.	ИП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
40.	×			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
41.	ИП			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
42.	M+			-		0
43.	СК			-		6 8 4 4 4 4 4 44 4 4 4 4 4 4 4 2
44.	СП			-		0
45.	'			-		0 , 0 I
46.	0I			-		0 , 0 I
47.	+			-		0
48.	=			-		0
49.				-		0